

Participant Weighted Knowledge Scores Report

Based on the calculated 1-p weights (inverse difficulty), a weighted knowledge score was computed for each of the 201 participants. For each participant, they received the 1-p value for every correct single-choice answer and every correct multiple-choice option they selected. Incorrect or “I don’t know” options contributed 0 to the score.

Summary Statistics

- **Total Possible Score:** 10.3284
- **Mean Score:** 3.2987
- **Median Score:** 2.8856
- **Standard Deviation:** 2.0377
- **Minimum Score:** 0.0000
- **Maximum Score:** 10.3284

Score Distribution

Metric	Value
Top 10% Cutoff	5.9055
75th Percentile (Q3)	4.3035
50th Percentile (Median)	2.8856
25th Percentile (Q1)	1.8806
Bottom 10% Cutoff	1.0547

A full list of participant scores has been exported to:
Participant_Weighted_Scores.csv

Item Weights Used

- **15. Is thalassemia a blood-related disease?** (Correct: Yes) -> Weight: 0.0995
- **** 17. What is the most severe form of thalassemia?**** (Correct: Thalassemia major (severe form)) -> Weight: 0.3731
- **19. Does thalassemia major require lifelong treatment?** (Correct: Yes) -> Weight: 0.3582
- **20. Can thalassemia major be cured?** (Correct: Very difficult (e.g., bone marrow transplant)) -> Weight: 0.8507
- **21. Can the spread of thalassemia be prevented?** (Correct: Can be prevented) -> Weight: 0.2488
- **22. How is thalassemia transmitted?** (Correct: From generation to generation (hereditary)) -> Weight: 0.1095
- **23. Is a thalassemia carrier usually sick or healthy?** (Correct:

Healthy) -> Weight: 0.1741

- **24. A child born from two thalassemia carriers will be:** (Correct: Has a chance to be affected (e.g., 25%)) -> Weight: 0.3980
- **26. How many thalassemia births occur in Sri Lanka per year?** (Correct: 40–100) -> Weight: 0.9602
- **16. What are the clinical form... / Thalassemia major (severe form) -**
-> Weight: 0.2687
- **16. What are the clinical form... / Thalassemia minor or trait (carrier form) -**
-> Weight: 0.2736
- **16. What are the clinical form... / Thalassemia intermedia (moderate form) -**
-> Weight: 0.9254
- **27. Problems faced by thalasse... / Regular blood transfusions for life -**
-> Weight: 0.1592
- **27. Problems faced by thalasse... / Iron overload affecting heart and liver -**
-> Weight: 0.7363
- **27. Problems faced by thalasse... / Delayed growth and puberty -**
-> Weight: 0.7612
- **27. Problems faced by thalasse... / Risk of diabetes, infertility, and hormonal issues -**
-> Weight: 0.9254
- **27. Problems faced by thalasse... / Risk of infections from transfusions -**
-> Weight: 0.8060
- **27. Problems faced by thalasse... / Fatigue -**
-> Weight: 0.6816
- **27. Problems faced by thalasse... / Bone deformities -**
-> Weight: 0.7960
- **28. What should a thalassemia ... / Get the partner tested before marriage -**
-> Weight: 0.2090
- **28. What should a thalassemia ... / Get family members tested -**
-> Weight: 0.2139